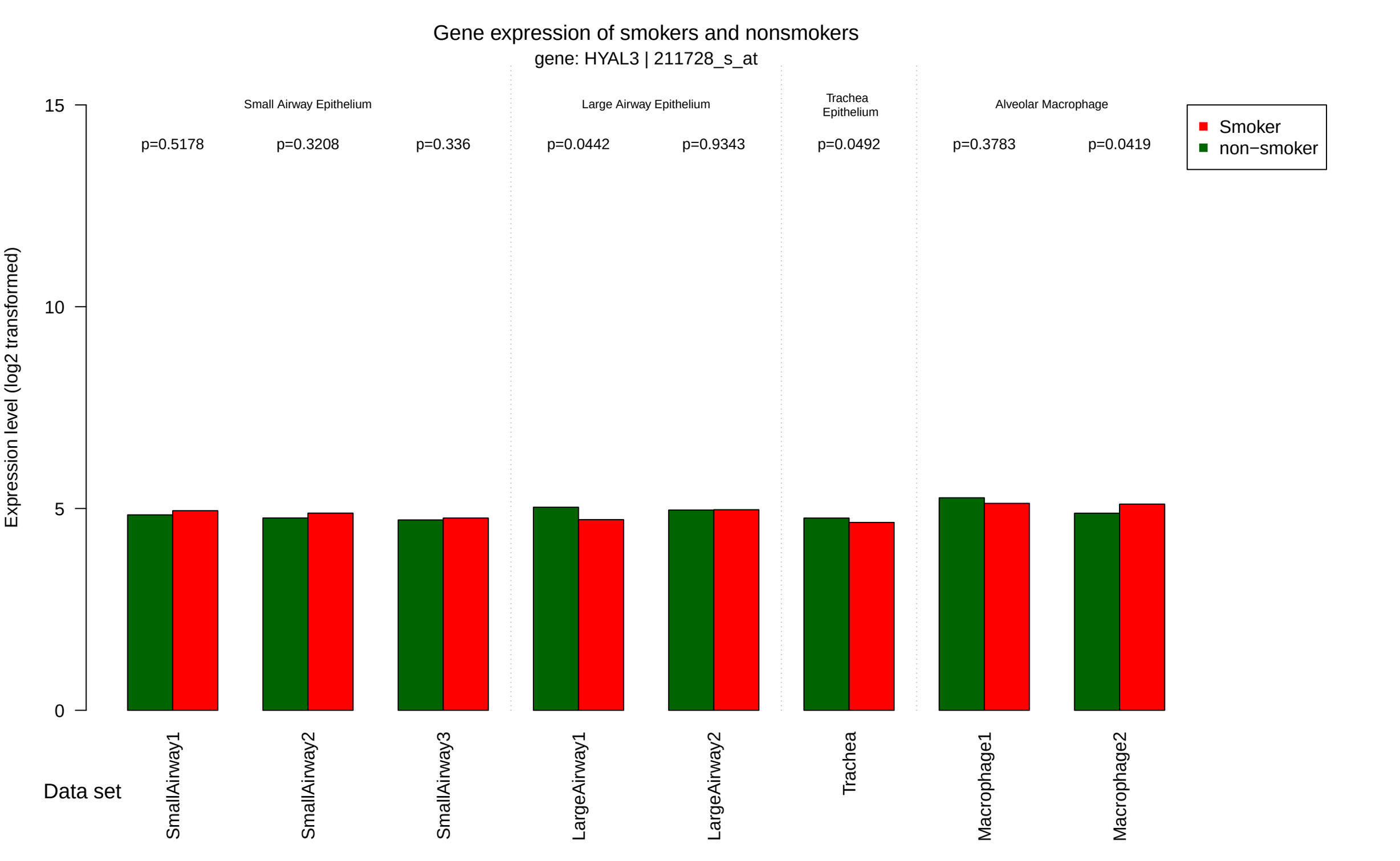
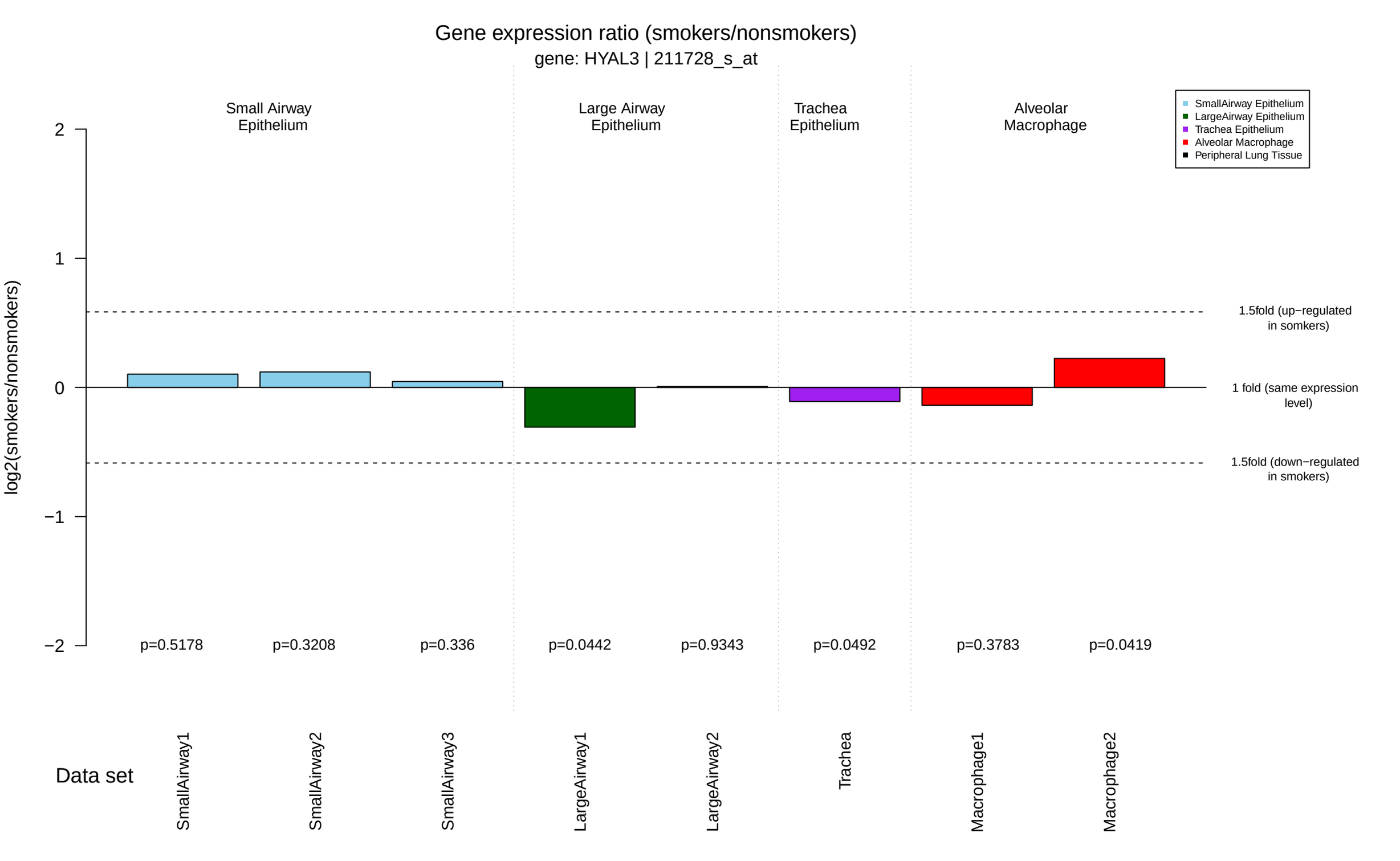
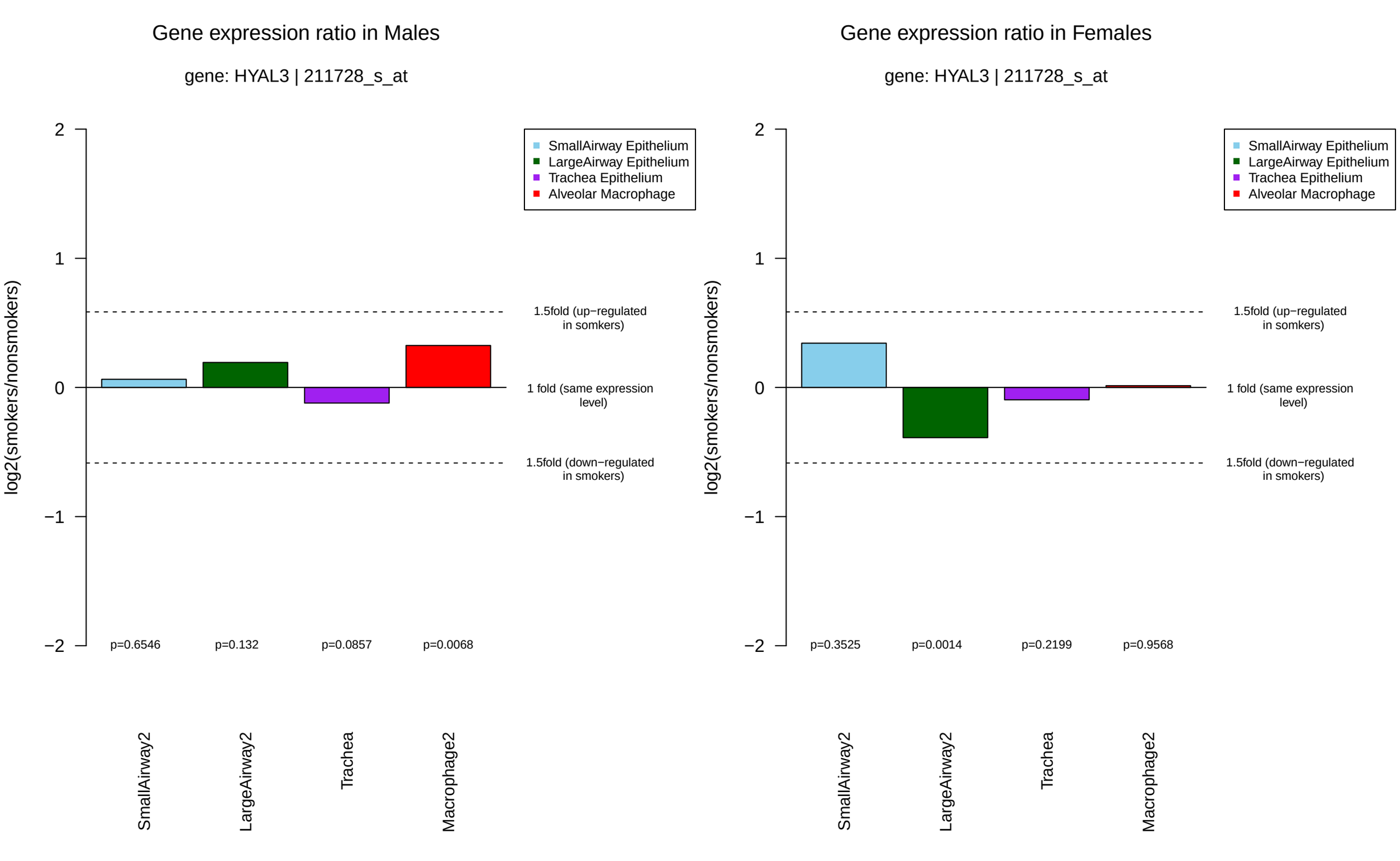
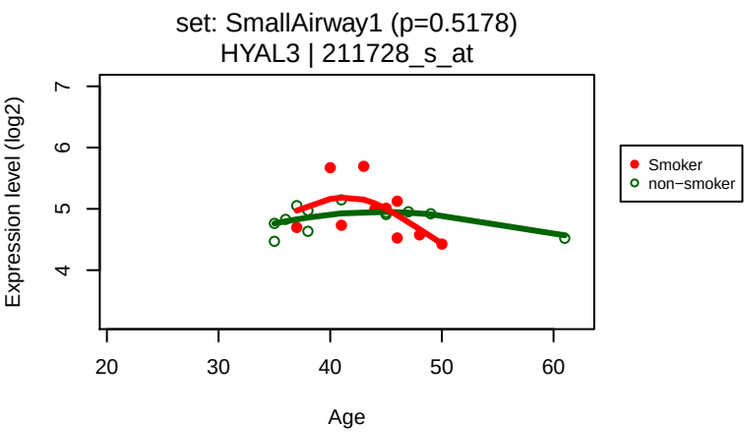
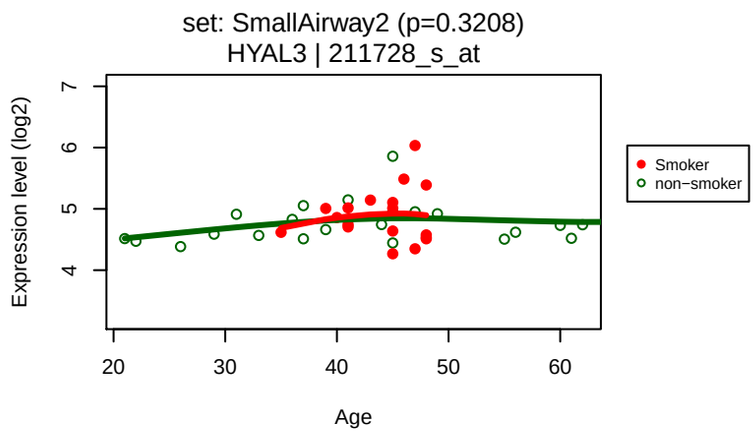
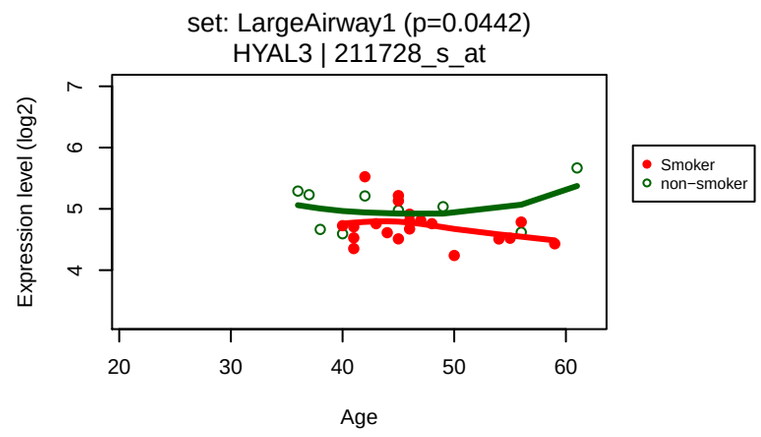

Gene expression of smokers and nonsmokers
gene: HYAL3 | 211728_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5178
p=0.3208
p=0.336
p=0.0442
p=0.9343
p=0.0492
p=0.3783
p=0.0419
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

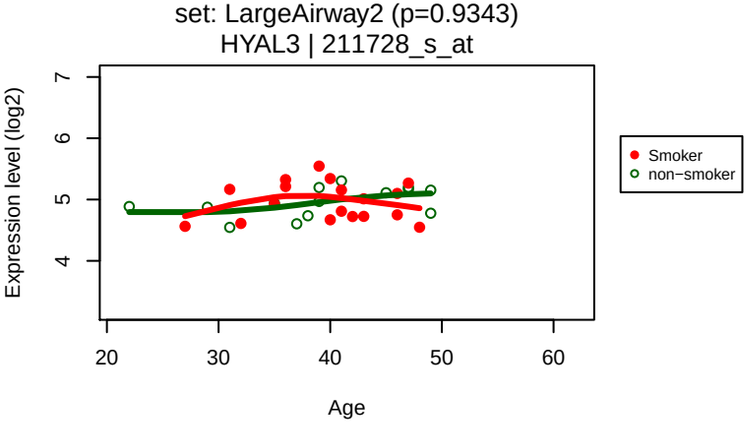

Gene expression ratio (smokers/nonsmokers)
gene: HYAL3 | 211728_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5178
p=0.3208
p=0.336
p=0.0442
p=0.9343
p=0.0492
p=0.3783
p=0.0419
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

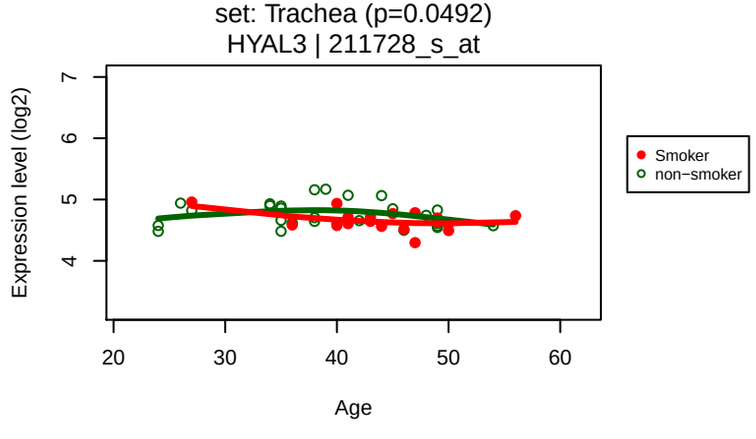

Gene expression ratio in Males
gene: HYAL3 | 211728_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6546
p=0.132
p=0.0857
p=0.0068
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HYAL3 | 211728_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3525
p=0.0014
p=0.2199
p=0.9568
SmallAirway2
LargeAirway2
Trachea
Macrophage2

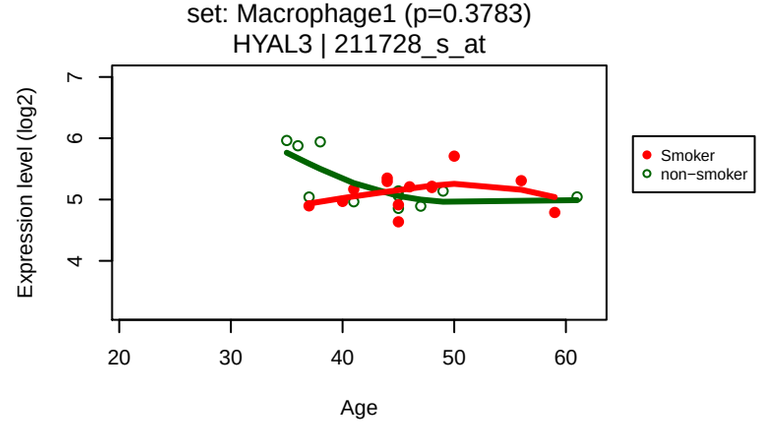

set: SmallAirway1 (p=0.5178)
HYAL3 | 211728_s_at
Expression level (log2)
Age
Smoker
non−smoker

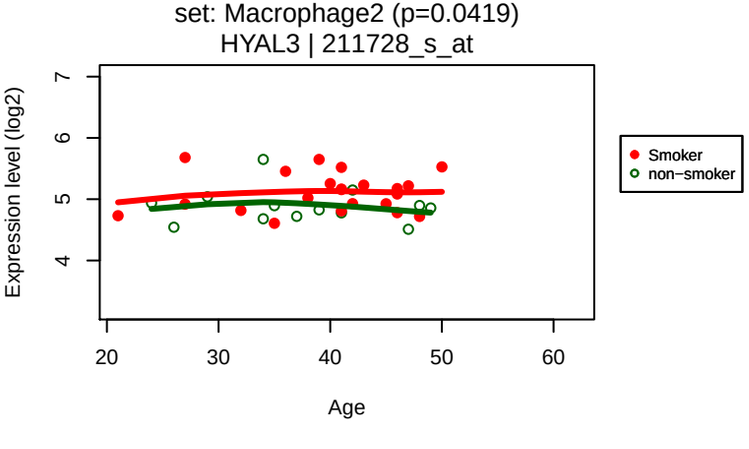

set: SmallAirway2 (p=0.3208)
HYAL3 | 211728_s_at
Expression level (log2)
Age
Smoker
non−smoker

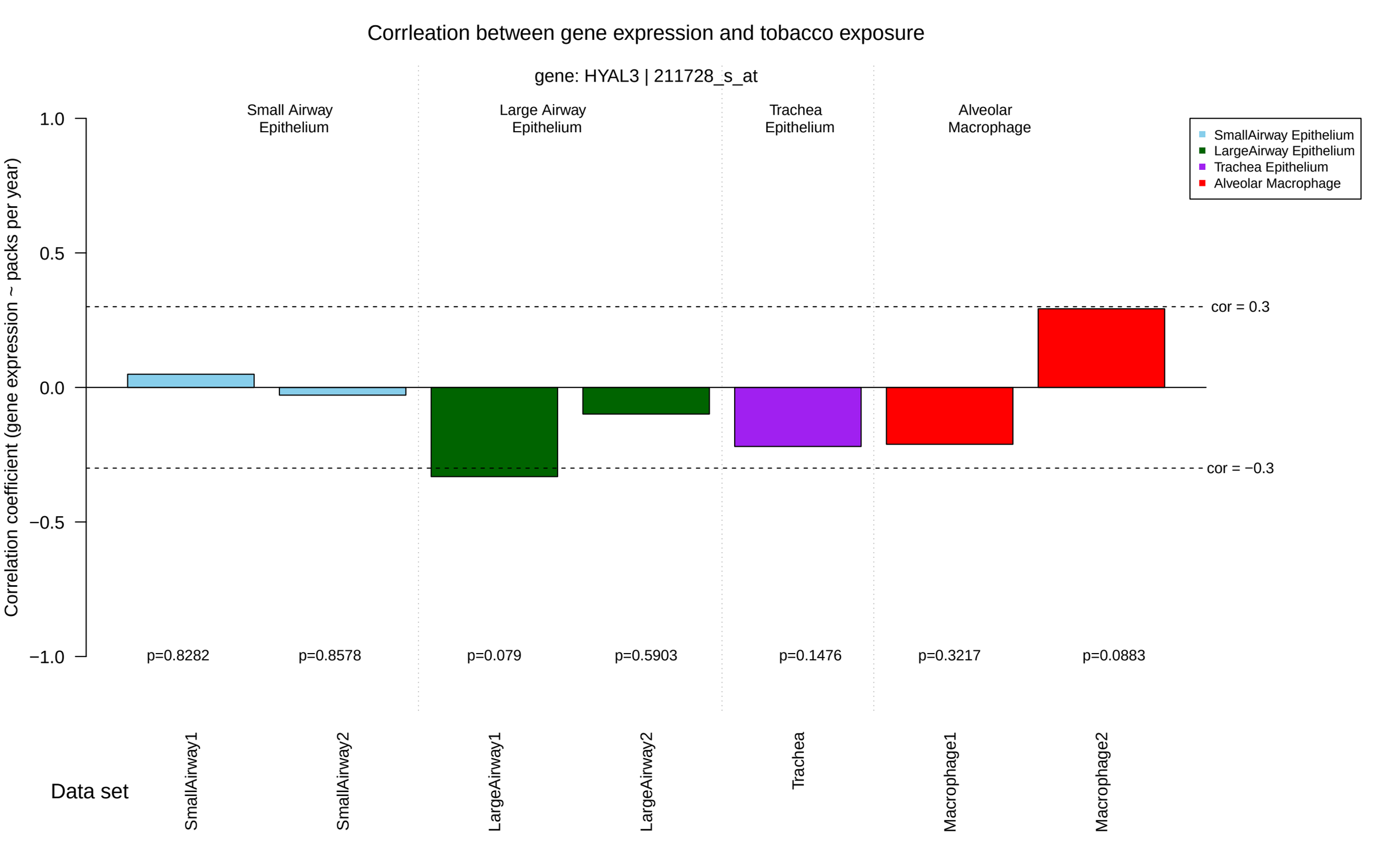

set: LargeAirway1 (p=0.0442)
HYAL3 | 211728_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9343)
HYAL3 | 211728_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0492)
HYAL3 | 211728_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3783)
HYAL3 | 211728_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0419)
HYAL3 | 211728_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HYAL3 | 211728_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8282
p=0.8578
p=0.079
p=0.5903
p=0.1476
p=0.3217
p=0.0883
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2